

D.V. 1

Taber Valt
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1) Pokaŕte, ŕe V komplexnŕm vektor. V se skalovnŕm soustavou pŕi.

$$a) \operatorname{Re}(\langle u, v \rangle) = \frac{1}{4} (\|u+v\|^2 - \|u-v\|^2)$$

$$= \frac{1}{4} (\|u+v\|^2 - \|u-v\|^2) \stackrel{\text{definice normy}}{=} \frac{1}{4} (\langle u+v, u+v \rangle - \langle u-v, u-v \rangle)$$

$$\stackrel{1)}{\|x\| = \sqrt{\langle x, x \rangle}} \stackrel{\text{linearity}}{=} \frac{1}{4} (\langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle -$$

$$- \langle u, u \rangle - \langle u, -v \rangle - \langle -v, u \rangle - \langle -v, -v \rangle) =$$

$$\stackrel{\text{linearity}}{=} \frac{1}{4} (\langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle + \langle u, v \rangle + \langle v, u \rangle -$$

$$- \langle v, v \rangle) = \frac{1}{4} (2\langle u, v \rangle + 2\langle v, u \rangle) =$$

$$\stackrel{\text{symetrie}}{=} \frac{1}{2} (\langle u, v \rangle + \overline{\langle u, v \rangle})$$

$$\text{Nechť } \langle u, v \rangle := a + bi, \text{ pak } \frac{1}{2} (\langle u, v \rangle + \overline{\langle u, v \rangle}) =$$

$$\in \mathbb{C} \quad a, b \in \mathbb{R}$$

$$= \frac{1}{2} (a + bi + a - bi) = \frac{1}{2} 2a =$$

$$= a = \operatorname{Re}(\langle u, v \rangle) \quad \square$$

$$\begin{aligned}
 b) \operatorname{Im}(\langle u, v \rangle) &= \frac{1}{4} (\|u - iv\|^2 - \|u + iv\|^2) \stackrel{\text{def. norm}}{=} \\
 &= \frac{1}{4} (\langle u - iv, u - iv \rangle - \langle u + iv, u + iv \rangle) \stackrel{\text{linearität + Skalarprodukt}}{=} \\
 &= \frac{1}{4} (\langle u, u \rangle - \langle u, iv \rangle - \langle iv, u \rangle + \langle iv, iv \rangle - \\
 &\quad - \langle u, u \rangle - \langle u, iv \rangle - \langle iv, u \rangle - \langle iv, iv \rangle) =
 \end{aligned}$$

$$\stackrel{\text{Skalarprodukt}}{=} \frac{1}{4} (-2 \langle u, iv \rangle - 2 \langle \overline{u}, iv \rangle) \stackrel{\text{linearität}}{=} -\frac{1}{2} (i \langle u, v \rangle - i \langle \overline{u}, \overline{v} \rangle)$$

$$\text{Hier } \langle u, v \rangle := a + bi \quad , \text{ mit } a, b \in \mathbb{R}$$

$$-\frac{1}{2} (i \langle a + bi, v \rangle - i \langle a - bi, v \rangle) = -\frac{1}{2} (ai - b - ai - b)$$

$$= b = \operatorname{Im}(\langle u, v \rangle) \quad \blacksquare$$